CHAPTER A

Reasoning and preparing for the exam

A.1 Logic: from words to proofs

Sets and statements

$x \in A$ means that x is an element of A; $A \subseteq B$ means that each element of A belongs to B. The empty set is denoted $\varnothing$. Intersection corresponds to "and", union to inclusive "or". In a universal set E, $E \setminus A$ is the complement of A. A pair $(a; b)$ is ordered; $A \times B$ groups the pairs whose first component is in A and second in B. The cardinality of a finite set is its number of elements.

Worked example. For $A = \{1; 2\}$ and $B = \{2; 3\}$, the intersection is $\{2\}$, the union $\{1; 2; 3\}$. The product $A \times B$ contains (1; 2) , $(1; 3), (2; 2)$, (2; 3): cardinality 4.

Numbers and intervals. $\mathbb{N}$ here contains 0 and the natural numbers; $\mathbb{Z}$ the integers; $\mathbb{D}$ the terminating decimals; $\mathbb{Q}$ the quotients of two integers, with non-zero denominator; $\mathbb{R}$ the reals. We have $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{D} \subset \mathbb{Q} \subset \mathbb{R}$. Thus $\frac{1}{3}$ is rational but not a decimal; $\sqrt{2}$ is real but not rational (earlier result). [-2; 3[means $-2 \leq x < 3$; an infinite endpoint is never included.

An implication "if P, then Q" expresses that P is a sufficient condition for Q and Q a necessary condition for P. Its converse is "if Q, then P". Its contrapositive is "if not Q, then not P"; it has the same truth value as the implication. An equivalence combines the two implications.

Worked example. "If x = 2, then x^2 = 4" is true. The converse is false for $x = -2$. The equality x = 2 is sufficient for x^2 = 4, but not necessary. The contrapositive "if $x^2 \neq 4$, then $x \neq 2$" is true.

To refute a universal statement, one counterexample is enough. To prove that an object exists, we can construct one and check its properties. Negating "for all x, P(x)" gives "there exists x such that not P(x)"; negating "there exists x, P(x)" gives "for all x, not P(x)". The symbols for quantifiers are not required; the meaning of the sentences is.

Worked example. The negation of "for every real x, $x^2 > 0$" is "there exists a real x such that $x^2 \leq 0$". The choice x=0 satisfies this negation.

Case analysis. Solve $|x|$ = 2 by distinguishing x positive or zero from x negative: we obtain 2 and -2. The cases cover the whole domain.

Contradiction. Suppose that x^2 + 1 = 0 had a real solution. We would have $x^2 = -1$, contradicting the positivity of a square. So there is no real solution.

Identity, equation, parameter. $(x + 1)^2 = x^2 + 2x$ + 1 is a true identity for all real x. x^2 = 1 is an equation whose unknown is x. In $x^2 - 2mx$ + 1 = 0, where m is a fixed parameter when solving for x; a case analysis on m is a different level of reasoning.

Cross-chapter exercises

1. Negate: "There exists a real x such that $x^2 < 0$". Is the original statement true?

2. Give the converse and the contrapositive of: "If an integer is a multiple of 4, then it is even". Which ones are true?

3. For $A = \{0; 1\}$ and $B = \{1; 2; 3\}$, calculate $A \cap B$, $A \cup B$, $A \times B$ and their cardinalities.

4. Show by contradiction that there is no largest natural number.

5. Classify -2, $0{,}125$, $\frac{1}{3}$, $\sqrt{2}$ in the smallest standard set; translate $x > 1$ and $x \leq 4$ as an interval. State the role of x and m in $mx + 1 = 0$. Solve according to the value of m.

Cross-chapter solutions

1. Negation: for all real x, $x^2 \geq 0$, true; the original statement is false.

2. Converse false, counterexample 2. Contrapositive: if an integer is not even, it is not a multiple of 4; true.

3. Intersection $\{1\}$, cardinality 1; union $\{0; 1; 2; 3\}$, cardinality 4; product of six pairs $(0; 1), (0; 2), (0; 3), (1; 1), (1; 2), (1; 3)$.

4. If N were the largest, N+1 would be a strictly larger natural number: contradiction.

5. Respectively $\mathbb{Z}$, $\mathbb{D}$, $\mathbb{Q}$, $\mathbb{R}$; interval]1; 4]. x is the unknown, m a parameter. If $m \neq 0$, a unique solution $-\frac{1}{m}$; if m=0, the equation becomes 1 = 0, no solution.

A.2 Python: understanding what is executed

The files provided run with Python 3. A function receives arguments and returns a result; a loop repeats an action. A program variable can change value by assignment, unlike a fixed unknown in an equation.

Lists and indices

`L=[1,4,9]` defines a list explicitly. `L.append(16)` adds one element; `del L[0]` removes the first. `L[0]` reads the element at index 0 and `len(L)` gives the length. A list comprehension, such as `[k*k for k in range(5)]`, contains the squares from 0 to 4. Order and repetitions appear in the list: it is not a set.

Worked example. After `L=[1,4,9]`, `L.append(16)`, then `del L[0]`, the list equals [4,9,16]. The instru `[x for x in L if x%2==0]` produces [4,16]. The test is a logical statement evaluated for each element.

File	Uses and checks to carry out
`suites.py`	Terms, sums, thresholds, factorial, Fibonacci, Syracuse, manipulations of lists. Test n=0, q=0, q=1, threshold already reached and threshold inaccessible.
`analyse.py`	Rates of change of secants, Newton, Euler, exponential threshold. Avoid zero step and zero derivative; distinguish stopping from convergence.
`probabilites.py`	Distribution, moments, Monte Carlo, samples, letter frequency. Check the sum of the probabilities, non-zero size and simulation seed.

`archimede.py`	Inscribed and circumscribed polygons, without using pi in the construction. Check that the bounds are tightening.
`verifier.py`	Run the deterministic checks and edge cases. A simulation close to a theoretical value does not constitute a proof of it.

Practical work to share out

1. Sequences: generate ten terms, compare loop and formula, then calculate a running total. Explain the length of the list.
2. Secants: take three positive and three negative steps; compare with the algebraic calculation of the rate.
3. Monte Carlo: compare 1,000 then 10,000 points with two seeds; comment on the fluctuation.
4. Samples: simulate N=1,000 samples of size n=10 then n=100; compare the observed mean and the expectation. Count the means within the threshold $\frac{2\sigma}{\sqrt{n}}$.
5. Euler and Archimedes: produce a table of results at several resolutions; state what we observe and what we have proved.

These practical sessions are alternatives spread over the algorithm and consolidation sessions. They are not all added to the same week of lessons. The results of a simulation depend on the seed; the check report gives an example of a reproducible run.

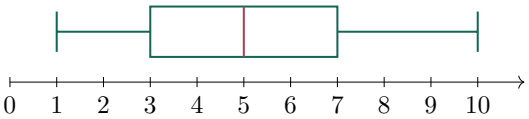
A.3 Reading and representing data

A weighted mean is $\frac{\sum n_i x_i}{\sum} n_i$. The median splits an ordered series; for a total count even, here we use the mean of the two middle values. The quartiles with ceiling ranks $\frac{n}{4}$ & $\frac{3n}{4}$ are explicitly used when values are asked for. The descriptive variance uses the denominator n, without estimation correction.

In a bar chart, the height represents the count or the frequency. In a histogram with classes of unequal widths, it is the area of the rectangle that must be proportional to it. A pie chart divides up the 360 ° according to the proportions. A box plot represents indicators of location and spread, not the frequency of each value. Always read the origin, units and scale before comparing.

Worked example. Two classes of width 2 and 4 each contain 20 observations. Heights 10 and 5 both give an area of 20. Equal heights would make the second class artificially twice as large.

Immediate application. Categories A, B, C have frequencies 10, 20, 30. Draw a bar chart and calculate the angles of a pie chart. For the series 1,1,3,3,7, give the mean, median, range and variance. Read the box plot below: minimum, quartiles, median and maximum. Can all the data be recovered?



Box independent of the previous series; whiskers at the extreme values.

Solution. Box: minimum 1, first quartile 3, median 5, third quartile 7, maximum 10. These five statistics do not determine all the observations. Total count 60: angles 60 °, 120°, 180°. The bar heights are 10,20,30. Series: mean 3; median 3; range 6; variance $\frac{4+4+0+0+16}{5} = \frac{24}{5}$.

A.4 Past papers: reading the format without narrowing the syllabus

The 2026 papers were consulted to identify the methods required. They fall under the first session of the EAM, before the 2026 curriculum applies; they serve as practice after checking the notions studied. The original EV papers are not presented as official papers.

Source	Methods observed	Reinvesting skills
Mainland France 2026, 26-MATSPEGEME1	Tree with an unknown probability; justified statements; geometry linking angle and equation.	C05, C03, C09; EV2-A, EV3-A and EV4-B.
Official paper 2026, 26-MATSPEGEG11	Discrete model, cumulative total and reading of loop; coordinates and angle; tangent and quotient.	C01, C03, C06; EV1-T1, EV2-T1.
Antilles-Guyane 2026, 26-MATSPEGEAG1	Threshold of a sequence; tree and random price ; exponential product and true/false.	C01, C05, C08, C09; EV4-T1 and EV4-B.
Éduscol specimen papers	Fluency, independent problems and written work without a calculator.	Practice T1/T2 for each batch.

The exact links are provided in the file `SOURCES.md`. The papers are not reproduced in full. To work on a past paper, note the paper's code, the exercise and the questions actually studied; correct the reasoning, not just the results.

Prepare a two-hour session without a calculator

Start by reading both parts; allow about 20 to 25 minutes for the fluency part. For an exercise, write the hypotheses and the domain before calculating. If you get stuck on a step, use the result given to carry on, saying so. Keep about ten minutes to check signs, the requested roundings, the units and the answers to the questions. These guidelines should be adapted to the pace observed during practice.